Received: 27 August 2023 | Accepted: 8 July 2024

DOI: 10.1111/1460-6984.13095

RESEARCH REPORT

International Journal of Language & Communication Disorders

Cross-neurotype communication from an autistic point of view: Insights on autistic Theory of Mind from a focus group study

Eleonora Marocchini[1] | **Irene Baldin**[2]

[1]Social Sciences Unit, Institute for Globally Distributed Open Research and Education, Gothenburg, Sweden

[2]Foundation COME Collaboration, Pescara, Italy

Correspondence
Eleonora Marocchini
Email: eleonora.marocchini@igdore.org

Abstract

Background: The conceptualisation of autism as a disorder where Theory of Mind (ToM) and pragmatics are fundamentally impaired has prompted a wealth of research on autistic deficits, most of which is characterised by two main assumptions: first, that autistic people would display said deficits, if present, with any conversation partner and in any situation; second, that neurotypical people do not present these deficits, regardless of the conversation partner. However, this is not necessarily reflected in autistic accounts of the way they experience social cognition and pragmatics.

Aims: The present paper aims to investigate the autistic experience of communication with both autistic and neurotypical people, with a particular focus on their perception of the ability of autistic and neurotypical people to understand their communicative intentions.

Methods & Procedures: Participants, 23 adult Italian autistic people without intellectual disability or language disorders, were recruited online. Two virtual focus groups of 2 hours each were conducted, transcribed and analysed through thematic analysis with a descriptive phenomenological approach by two independent researchers.

Outcomes & Results: Six themes were developed from the analysis, the most relevant being Autistic–Autistic communication and Autistic ToM. The results, in line with the Double Empathy theory, suggest there seem to be important differences between neurotypical and autistic people's ToM. These appear to make it easier for autistic people to communicate with one another, as well as to create difficulties for neurotypical people to understand autistic people, not just the other way around.

Conclusions & Implications: These results seem to confirm that challenges in cross-neurotype communication are better interpreted as mutual miscomprehension and reciprocal differences in ToM rather than deficits on the autistic part. This calls for a reframing of ToM and/or the need for autistic ToM as a construct, of which neurotypical people seem to be lacking. Moreover, these insights should be taken into account for speech and language therapy and clinical practice in general, advocating for a neurodiversity-informed view of co-constructed communication as well as for a broader societal change in which therapists can play a crucial role, through participatory approaches or raising awareness in their daily practice.

KEYWORDS
autism, communication, cross-neurotype communication, pragmatics, Theory of Mind

WHAT THIS PAPER ADDS

What is already known on the subject

- Autism is conceived as characterised by social cognition and communication difficulties, often linked to Theory of Mind (ToM) deficits. However, recent research suggested variations in ToM abilities within the autistic population and proposed alternative theories like the Double Empathy theory. Nevertheless, only a few studies examined how autistic individuals perceive communication across neurotypes.

What this study adds

- Autistic individuals seem to find it easier to communicate with other autistic people, and they identify specific characteristics of neurotypical communication that hinder successful communication. Moreover, neurotypical people are perceived as having difficulties in autistic ToM, which seems to emerge as a relevant and needed construct in light of the Double Empathy problem.

What are the clinical implications of this work?

- These findings can inform speech and language therapy and clinical practice about the potential gains of raising awareness on the Double Empathy problem and the higher communication ease inside the autistic community, alongside individualised support. Participatory approaches and closer collaboration with the autistic community also seem to be crucial for therapists to help improve communication experiences for autistic individuals.

INTRODUCTION

Autism is currently conceived as a disorder where social cognition and communication are impaired, as stated in the criteria of the American Psychiatric Association's diagnostic manual (APA, 2022), used worldwide. These impairments would manifest as deficits in verbal and non-verbal communication and in social-emotional reciprocity, failure to initiate and respond to social interaction. Most of these pertain to the scope of pragmatics, defined as a set of abilities to use language (and other paralinguistic means) to communicate in a specific context and to infer the speaker's meaning beyond the literal meaning. These abilities are closely linked to (or comprised in) social communication (Cummings, 2009). The role of intention recognition in pragmatics is so central that it has been framed in some literature (Sperber & Wilson, 2002) as a sub-module of Theory of Mind (ToM), a crucial construct that is part of social cognition. In fact, ToM has been defined as the ability to attribute mental states to oneself and others, to predict and explain their behaviour (Premack & Woodruff, 1978).

This conceptualisation of pragmatics as emerging from ToM prompted a wealth of research on ToM impairments as the potential cause for pragmatic deficits, particularly in autism (Gernsbacher & Pripas-Kapit, 2012; Marocchini, 2023), 'the archetypal disorder of social cognition' (Happé et al., 2017). Early formulations of the influential 'ToM deficit hypothesis' proposed a complete lack of mental states consideration in autism (Baron-Cohen et al., 1985; Frith, 1989) and framed it as the cause for various autistic behaviours. The theory shaped decades of research in pragmatics, partially explaining observed difficulties in recognising violations of maxims of conversation, that is, redundancy avoidance, informativeness, truthfulness, relevance and politeness (Surian, 1996) and pragmatic phenomena such as indirect requests (Paul & Cohen, 1985), metaphor and irony (Happé, 1993) and humour (Ozonoff & Miller, 1996).

The relationship between ToM and pragmatics is now known to be dependent on task, population, cognitive and linguistic abilities (Bosco et al., 2018). Moreover, theoretical and methodological issues involving ToM and empathy as constructs are increasingly discussed (Quesque & Rossetti, 2020; Schaafsma et al., 2015). Affective empathy and ToM are often defined as the 'affective and cognitive routes' (Kanske, 2018) to sharing other people's feelings and emotional states (whilst knowing they are not ours) and comprehending their intentions and mental states (Stietz et al., 2019). Although these routes can function independently (Kanske et al., 2016), 'empathy' is sometimes used as an umbrella term for both, and tests that conflate the two—such as the Empathy Quotient test (Baron-Cohen & Wheelwright, 2004)—are often used in autism research, feeding into the misconception that autistic people have generalised empathy deficits. Other commonly used tests show poor internal consistency (Olderbak et al., 2015) and questionable removal of items to highlight autistic differences (Chevallier, 2012).

Epistemic disparities emerge comparing pragmatic research on the general population, emphasising co-constructed meanings and individual differences, and clinical pragmatics, using tasks with an 'accurate' response, and norming studies on neurotypical individuals (Marocchini, 2023). Different epistemic postures can also be observed in results interpretation: when no deficit is found, researchers argue that surface-level performance might not mean competence, and frame autistic strategies as 'compensatory' (e.g., Livingston et al., 2019, Corbett et al., 2021), presuming the existence of a singular (and superior) route to comprehension.

Currently, we have both early and recent studies suggesting ToM abilities can develop later in life (Happé, 1995; Scheeren et al., 2013), are not universally impaired in autism (Loukusa et al., 2023) and are less relevant than IQ and linguistic abilities to account for autistic difficulties (Gernsbacher & Yergeau, 2019). Similarly, several studies found no difficulties in autistic people with pragmatic phenomena such as lexical ambiguity resolution, scalar implicatures, indirect requests and even deception (e.g., Brock et al., 2008; Chevallier et al., 2010; Marocchini et al., 2022; van Tiel et al., 2021) and mixed evidence with regard to politeness. In fact, some studies report that autistic people can adjust their politeness levels in familiar settings or when properly instructed (Sirota, 2004; Volden & Sorenson, 2009), others, that autistic individuals are perceived as more polite than neurotypical peers when in leading roles and less when in equal roles (Yang et al., 2021), suggesting differences might depend on conflicting views on what politeness is and when it is needed.

Pragmatics and ToM research on autism tend to disregard alternative paradigms and theories, renowned within autistic communities, such as the neurodiversity paradigm (Chapman, 2019) and the Double Empathy theory (Milton, 2012). The neurodiversity paradigm, born and appreciated within the autistic community, proposes a revolution in the framing of disorders. It posits that every brain is unique and characterised by different traits, challenges and abilities, and that no brain is right or wrong; moreover, neurodiversity includes everyone (Chapman, 2019). The neurodiversity paradigm describes the world (and clinical knowledge) as built according to the way neurotypical brains work; as such, it reframes autistic difficulties as stemming from a contrast between their characteristics and society expecting them to have the same needs and behaviours of neurotypical people, challenging clinical research and goals (Dawson et al., 2022). In fact, neurodiversity advocates point out the neuronormative, gender and racial biases in diagnostic and therapeutic practice (Marocchini, 2023). The Double Empathy theory, in line with this paradigm, challenges the perception of ToM deficits in autism, suggesting it could stem from a cross-neurotype communication issue, that is, a mismatch between autistic characteristics and neurotypical expectations. It posits that autistic people exhibit higher communication skills when interacting with autistic peers; conversely, neurotypical people could struggle in communicating with autistic people.

Recent empirical research offers support to this theory. Brewer et al. (2016) asked autistic and neurotypical participants to recognise emotions expressed by autistic and neurotypical posers. The findings highlight that neurotypical individuals had difficulties recognising emotions of autistic posers. Edey et al. (2016), asked autistic and neurotypical adults to generate animations portraying specific mental state interactions by directing two triangles and then to rate to what extent the animations generated by others depicted each of the target mental states. Neurotypical participants showed difficulties in inferring mental states depicted by autistic individuals.

These difficulties impact neurotypical-autistic reciprocity of communication styles, as suggested by participants' rating of their cross-neurotype interactions (Crompton et al., 2020c), as well as pragmatic mutual (mis)understanding: Williams et al. (2021) performed a relevance-theoretic analysis of conversations between dyads of autistic people conversing with autistic and non-autistic strangers and found that mutual understanding was unexpectedly high in autistic–autistic dyads.

However, research considering autistic perspectives on communication with conversation partners of different neurotypes remains underexplored. A good recent example is a qualitative study by Crompton et al. (2020a) on autistic experiences of spending social time with neurotypical and autistic friends and family: through semi-structured interviews, they highlighted the benefits of relationships between autistic people in terms of cross-neurotype understanding.

Our exploratory study aims to investigate autistic perceptions of cross-neurotype communication with a specific focus on ToM and indirect communication, in order to reduce the gap between traditional views stemming from clinical research and insights from these recent studies. Our two main research questions are:

1. Is communication perceived differently by autistic individuals depending on the interlocutor's neurotype (neurotypical versus autistic)?
2. How good are neurotypical people at understanding autistic people, according to autistic individuals?

These questions were operationalised into dimensions that could be investigated through a focus group study and subsequent qualitative analysis, to inform the field on the autistic perspective on the matter.

Positionality statement

The first author started researching pragmatics in autism during her PhD programme. Then, she found autistic communities online questioning pathologising accounts of autism. She self-identified as autistic, got formally diagnosed, and started engaging in participatory research. Given her training in the cognitive sciences, she still fights against positivist and stigmatising approaches to clinical data. She believes active reflexivity is needed in social and clinical sciences and engages in self-reflection on the ways doing research as a 'neurotypical' and a neurodivergent researcher have impacted and impacts her research.

The second author has conducted qualitative research in osteopathy. She self-identified as autistic and sought an expert opinion, who confirmed her suspicions. Meanwhile, she realised how significantly the clinical and autistic accounts differed, particularly regarding women's experiences of misdiagnosis and the pathologising approach expressed throughout her training. Thus, she has not yet sought a formal diagnosis in the healthcare system, and she still struggles to find and promote a more humane discourse on autism in clinical settings through self-reflective activity on how she might have internalised this bias through her training.

Although most autism research is conducted by neurotypical researchers and their reasoning is ingrained in our approach, we had neurotypical assistants leave notes and debrief with us after data collection to account for the potential bias we might introduce.

MATERIALS AND METHODS

The study was designed from a moderate social constructionist epistemological perspective. Social constructionism conceives social phenomena as constructed through discourse and interaction in a given context (where personal perspectives are rooted), subsequently internalised as part of people's world views (Burr, 1995; White, 2004). However, social constructs become socially accepted through power-imbalanced mechanisms (Burr, 1995). Thus, critical analysis of shared assumptions (White, 2004) and coherence between epistemology, methodology and method is encouraged—preferring qualitative ones, to capture unstructured accounts of personal views (Rees et al., 2020)

In line with this lens, we adopted a descriptive phenomenological approach (Giorgi, 1997; Pietkiewicz & Smith, 2014) with thematic analysis (Vaismoradi et al., 2013), known as a valid method to understand people's personal experience, going beyond long-held assumptions, as well as querying traditional knowledge, thus supporting the creation of new theories and responses (Giorgi, 1997).

As argued earlier, power imbalance between (often neurotypical) researchers and participants is prominent in autism studies and can be seen even in participatory research, as autistic people's accounts of their experience are usually not valued enough (Fletcher-Watson et al. 2019). This study was designed following this approach to critically analyse assumptions on social interaction constructed through the neurotypical lens.

The Standards for Reporting Qualitative Research checklist (Tong et al., 2007) was used to ensure consistency of the study methods and development.

Participants

Inclusion criteria stipulated that participants should be autistic, over the age of 18, without intellectual disability or language disorders, and Italian native speakers. Twenty-three adults (16 women, 1 man, 4 non-binary and 2 gender-questioning people[1]) between 19 and 52 years of age (mean age: 32.91; SD: 8.88), participated in the study. Participants were all White, with a diverse education, ranging from 13 to 21 years of study (mean: 16.09; SD: 2.23). Detailed demographic data are available on the Open Science Framework: https://osf.io/p8ubr.

In line with the neurodiversity paradigm, exposing biases and delays in diagnostic practice, we allowed participants to self-identify as autistic. They were all either formally diagnosed ($n = 20$) or self-identified ($n = 3$) as autistic in adulthood and were recruited via social media. All of them had engaged with neurodiversity-related content, such as social media or blog posts and online encounters on autism as neurodivergence and the neurodiversity paradigm.

Written informed consent was obtained from all participants. The research has been conducted in adherence to the Code of Research Conduct of the Institute for Globally Distributed Open Research and Education, in full compliance with the standards of APA ethical guidelines, and the 1964 Helsinki Declaration. The research protocol was approved by the COME Collaboration Institutional Review Board (IRB protocol #36/2022).

Materials

Data collection consisted of two Virtual Focus Group Discussions (VFGDs). Focus groups were chosen for two reasons: first, they allow for a rapid collection of diverse information (Morgan, 1996); second, they entail explaining oneself to other participants (unlike in individual interviews), enabling researchers to estimate consensus in a group (Carey & Smith 1994). Moreover, VFGDs support real-time communication across various locations (Tuttas, 2015) and do not require changing environments or enduring higher sensory stimulation, allowing for the examination of social dynamics between autistic participants, whilst preserving part of their sensitivities.

Three dimensions were derived by applying the neurodiversity lens to traditional literature on autistic communication: two main dimensions focusing on autistic deficits in pragmatics and ToM and a minor, exploratory, dimension focusing on politeness. To subvert the classic epistemic posture, the dimensions were reframed through insights from the Double Empathy theory, that is, as relational rather than individual issues. The questions were designed by an autistic researcher with expertise in pragmatics and ToM and checked for clarity by two neurotypical qualitative methods experts: they were phrased not to name any investigated construct nor bias participants towards specific answers (see Table 1). They were always asked the same way by the same researcher.

First, are difficulties observed in autistic communication pragmatic deficits or effects of mutual misunderstanding due to neurotype differences? If there were deficits in autistic communication as a whole, autistic people should experience the same difficulties with neurotypical and autistic interlocutors. Second, are autistic challenges in understanding others' intentions a general ToM impairment or effects of ToM differences between neurotypes? If autistic people had impaired ToM abilities and neurotypical people had good ToM abilities, neurotypical people should have no difficulties interpreting autistic people's intentions. Lastly, are autistic people incapable of managing politeness or is politeness particularly prone to neuronormative standards that are not necessarily shared? The exploratory question was broad but referred to the two extremes to address the mixed evidence in the literature, as autistic people can be perceived as both rude and excessively polite.

The prompts also included more specific questions on small talk, indirect communication, and ways of displaying attention. However, participants talked about these spontaneously, so they were not asked. Data saturation was reached after the second VFGD, without new themes being elaborated.

Procedure

Two VFGDs, each lasting 2 h, were conducted via Zoom by two facilitators: an autistic moderator and a neurotypical assistant. A total of 10 and 13 participants took part in the two VFGDs, respectively. They started with rules for discussion and a round of warmup presentations of each person's special interests. Participants consented again through Zoom's pop-up consent to video-record.

Following previous literature, we used a more structured approach to ensure equal participation (Morgan, 1996), as some autistic participants might not be able to join the conversation or, conversely, engage in autistic monologues and turn-taking overlaps (see Ying et al., 2018). Moreover, recent studies suggest video-mediated interactions can increase turn-taking problems, with latency disrupting interactions without participants recognising them as technological issues (Seuren et al., 2021). Zoom's audio selection feature (suppressing overlapping speakers' audio) enhances these issues.[2] Several measures were taken to circumvent them and adapt the methodology to autistic functioning. Participants were instructed to raise their digital hand in Zoom when they felt ready to respond to the question. They could also signal (dis)agreement with others' statements both physically and through

TABLE 1 Dimensions and questions.

Dimensions	Questions
D1: Autistic deficits in comprehension or mutual misunderstanding?	Q1: 'Do you find any difference in communication with neurotypical people or autistic people like you (when you know about the other person's neurotype)? If you do, what are they?'
D2: ToM impairment in autism or different ToM between neurotypes?	Q2: 'It is often said that part of communication is reading other's intentions and 'minds'. In your experience, are neurotypical people good at understanding yours?'
D3: Autistic lack of politeness or neurotypical normativity in politeness management?	Q3: 'How is your communication style usually perceived? Does anybody tell you that you are excessively polite or rude?'

digital reactions, as well as using the chat. The moderator would report these contributions verbally, to incorporate them in the conversation and transcription. Participants could also turn their cameras off to stim (i.e., engage in self-stimulatory or self-soothing behaviors) or not to be seen for a while. The assistant would privately tell the moderator if someone was not responding to one of the questions or was talking for more than 3 min, so that the moderator would encourage whoever had not spoken yet to intervene. An optional feedback questionnaire was sent to evaluate how comfortable participants felt and provide an opportunity to express additional remarks.

A 30-min debriefing occurred immediately after both VGFDs: the neurotypical assistant shared field notes and provided his perspective. Transcriptions followed a subset of CLIPS protocols for Italian dialogic data (Savy, 2007), participants were anonymised through numerical tags. Before the analysis, transcriptions were sent for confirmation: none of the participants suggested any change.

Thematic analysis

Transcripts were inductively analysed by two researchers, following Giorgi's methodological steps (1997) for descriptive phenomenological analysis. Field notes were used to support thematic analysis (Vaismoradi et al., 2013). To enhance data credibility, we adopted several methodological strategies, following Graneheim and Lundman (2004). Before starting the analysis, we discussed our potential personal impact as autistic researchers; reflexivity played an important role in reducing the risk of disregarding group effects and potential conformity/censorship issues in data collection and analysis (Carey & Smith, 1994) through reiterative reading and discussion of personal views and biases, thus enhancing confirmability (Rees et al., 2020).

We independently analysed the transcriptions tagging them in Taguette 1.4.1 (©Rémi Rampin). Each author thoroughly read the transcriptions and identified relevant parts through codes. These were debated in peer debriefings through analysts' triangulation, to increase data consistency (Patton, 1999; Leung, 2015). After this process, codes were grouped into abstract categories and themes through conceptual analysis in a preliminary codebook. Then, we thoroughly discussed categories and themes, reached consensus over controversies, merged and renamed overlapping categories, and kept those that seemed relatively distinct, even when conceptually connected. A particularly wide theme was redefined into two more precise themes (*Neurotypical ToM* and *Autistic ToM*). After various meetings and in-depth analysis in a shared process, we outlined the final codebook and generated the thematic map. The analysis was presented to participants for validation.

RESULTS

Twenty-three people participated in the study, with no dropouts, and 18 filled the feedback questionnaire. In terms of group-level dynamics, field notes and debriefings suggested that the groups worked well, introducing interesting topics, from a novel perspective. Many participants expressed agreement in the chat or nodding; several mentioned taking notes of talking points to continue the discussion only when they felt like intervening. All participants expressed themselves multiple times and thanked everyone for the time spent together. Feedback questionnaires report a positive atmosphere and three additional remarks: on the potential impact of early diagnosis ($n = 1$) and on the moderator's ability to make space for everyone ($n = 2$).

Three overarching themes were developed from the analysis, each encompassing two themes. Each of these six themes included 2–8 categories or sub-themes (see Table 2).

The first overarching theme deals with *One's Communication Style*, depending on one's neurotype (not the interlocutor's), and includes *Autistic Communication* and *Neurotypical Communication* themes. *Interpersonal Communication*, the second overarching theme, depends on both one's neurotype and their interlocutor's,

TABLE 2 Themes and categories.

Overarching themes	
One's communication style	
Themes	**Categories**
Autistic communication	Directness
	Autistic monologue
	Special interests
	Energy demand
	Sensory interferences
	Masking and camouflaging
	Higher ease with children
Neurotypical communication	Indirectness
	Missing information
	Small talk
	Neurotypical scripts
	Neurotypical masking
Interpersonal communication	
Autistic–Neurotypical communication	Turn-taking
	Rage due to misunderstanding
	Hypervigilance
	Over-politeness
Autistic–Autistic communication	Higher comprehension
	Quality of the conversation
	Higher ease
	Unmasking
Cognition in communication	
Neurotypical ToM	Autistic difficulties in neurotypical ToM
	Differences in non-verbal communication
Autistic ToM	Neurotypical difficulties in autistic ToM
	Neurotypical overinterpretation

Abbreviation: ToM, Theory of Mind.

covering *Autistic–Autistic Communication* and *Autistic–Neurotypical Communication* themes. The third overarching theme regards *Cognition in Communication*, particularly ToM, comprising *Neurotypical ToM* and *Autistic ToM* themes.

Themes are exemplified through brief descriptions and representative quotes (translated into English) per category. Some quotes are edited for clarity: missing parts of the quote are represented by '[…].' Other tags in '<…>' indicate short pauses (<sp>), laughs and hesitations (<eeeh>). A wider selection of unedited quotes can be found in the Supplementary Materials.

One's communication style

Differences in communication style between autistic and neurotypical people were developed as perceived by participants. These, represented in the themes *Autistic Communication* and *Neurotypical Communication*, are reported as core characteristics of one's communication, regardless of the interlocutor's neurotype (Figure 1).

Autistic communication

This theme considers the essence of autistic communication, including tendencies towards directness, monologues and special interests talk, and their experiences in terms of energy demands, sensory interferences, camouflaging and ease of talking with children.

Directness

Participants suggested most autistic people tend to be particularly direct, even blunt, if they do not camouflage. Some describe their directness as talking spontaneously,

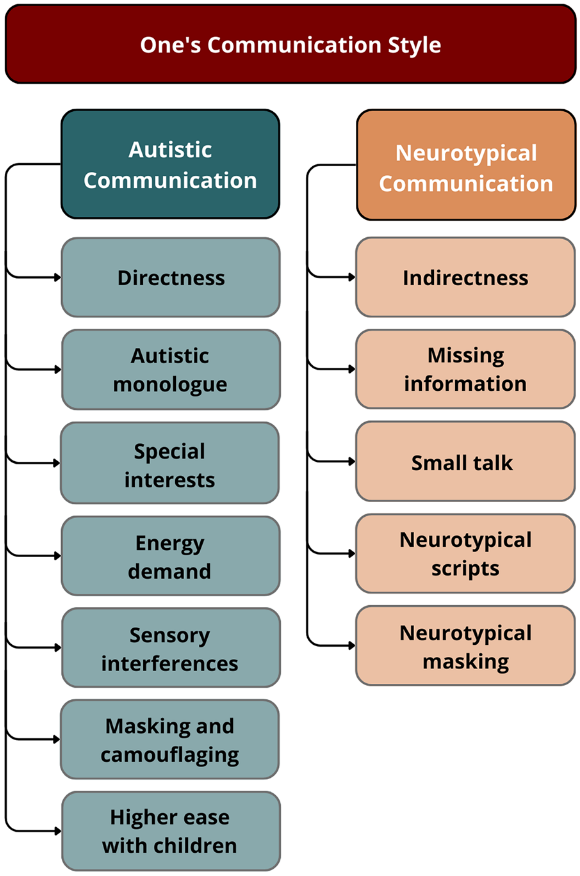

FIGURE 1 Themes and categories in one's communication style.

without filters, and assertively. They also reported difficulties understanding why someone would not say what they actually mean, like them. Some added they can be perceived as offensive, whilst just 'stating facts':

> *'for example, I simply state facts… but […] I notice I offend <sp> the people […] who love you endure, but you hurt them' (P03, diagnosed autistic woman, aged 37)*

Some participants find this tendency in autistic peers refreshing and comforting, as they know their interlocutors are plainly expressing their thoughts and they are not required to discover hidden meanings.

Autistic monologue

Participants reported monologue as typical of their communication, as a means to explain everything well, give depth of details and add more when they feel like they have overlooked something. They also reported disappointment for interruptions of their monologues, and they seem to envision a conversation made of monologues as preferable.

> *'in general, I don't like [being interrupted in] a monologue, that is, I listen to others' monologue, and then I'd like to have my turn <sp> if / if I could have a conversation in this way, it would be ideal for me' (P05, diagnosed autistic gender-questioning person, aged 38)*

Special interests

This category was closely linked to *Autistic monologue*, as autistic people tend to exhaustively present their special interest, resulting in monologues. Participants shared they tend to befriend people who talk about their interests (regardless of their neurotype), because then they can enjoy the time spent discussing and doing what they love. Moreover, people's interests are ways to better understand them, and to perceive them as more interesting.

> *'during the years, I've realised that all my strong friendships are based on special interests <sp> that is, we looked for each other based on our shared special interests <sp> because they keep us spending time together pleasantly' (P08, diagnosed autistic woman, aged 31)*

Talking with autistic peers is perceived as easier than conversing with neurotypical people because of special interests. Indeed, some have difficulties maintaining a conversation with neurotypical people without in-depth interests, and others were bullied by neurotypical schoolmates for their special interests.

Energy demand

Participants reported tiredness, even exhaustion, after social interactions. Several factors, such as masking, small talk, hypervigilance and interlocutors influence the amount of energy required. Indeed, cross-neurotype interaction seems to require more energy. This appears related to the need to control their voice and face expression when conversing with neurotypical people (see *Hypervigilance* in *Autistic–Neurotypical Communication*).

> *'I usually tend to invest plenty of energy <sp> and time in a conversation. I mean, if I'm not sure we have a certain amount of time and attention to dedicate to the conversation, I usually prefer not to start it' (P04, self-identified autistic woman, aged 25)*

Sensory interferences

Participants reported difficulties in conversing with a noisy background or with several interlocutors, even worse when they talk over one another, making it difficult to hear properly and to stay focused. Sensory interferences are distracting and tiring for autistic people, so this concept is linked to *Energy demand*. Reports primarily focused on auditory integration and eye contact, as well as internal stimuli such as hunger or thirst.

> *'I can't look people in the eyes, for me, it's, that's another thing I have to do <eeeh> that distracts me from what I have to say <sp> from remaining on the concept I want to express' (P08, diagnosed autistic woman, aged 31)*

Masking and camouflaging

Participants recalled learning to communicate in the normative 'proper' manner, as adults explained directness was impolite. Thus, they developed several means of mingling in the neurotypical society, such as people-pleasing. These strategies are so ingrained they employ them automatically, regardless of the interlocutor's neurotype. However, it is easier to abandon these strategies with other autistic people (see *Unmasking* in *Autistic–Autistic Communication*). Participants also reported being confused when someone behaved the way they were reprimanded for.

Others felt like 'speaking a different language' and decided to study communication, read articles and books and attend theatre courses. For some, it resulted in a highly empathetic communication, obtained through years of learning, performing and interpreting social cues.

> *'I still suffer, but less <sp> I mean, I fake it much better <laugh> <eeh> this understanding <eeh> of the implied language <sp> in fact, I've read a lot <laugh> about this and I've <sp> little by little, educated myself just as learning a new language' (P19, diagnosed autistic woman, aged 19)*

Masking and camouflaging are presented as survival tools, but they require great amounts of energy, making it difficult, even painful, to perform them sometimes. They also entail confusion about one's personality: some participants said they could not recognise themselves, after years of camouflaging.

> *'when I'm tired, I tend to be more autistic <laugh> I mean that <sp> the mask is more difficult to wear' (P08, diagnosed autistic woman, aged 31)*

Higher ease with children

Participants reported higher comprehension and ease whilst speaking with children, because they talk less 'artificially' than adults. Some also feel more comfortable and allowed to unmask, speak directly and stim, with them. Moreover, this seems to happen because autistic people perceive children to be careless about characteristics that neurotypical adults would perceive as weird or inappropriate.

> *'I find it less complicated with middle school children […] because to me they seem still 'uncontaminated' - it's a strange word <sp> but […] they're more genuine <sp> than adults' (P11, diagnosed autistic woman, aged 37)*

Neurotypical communication

This theme addresses the autistic perception of neurotypical communication, including their indirect communication style, encompassing tacit content and missing information. It also includes their use of small talk and what is perceived as neurotypical scripts and masking.

Indirectness

Participants highlighted the presence of implicit messages in neurotypical communication, often perceived through contrasts between body language and words and their difficulty understanding the real intentions of neurotypical interlocutors, for instance, irony and sarcasm, presuppositions and implied messages. Autistic people tend to consider the explicit, not the implicit message. Moreover, participants reported having to ask many questions to be sure they are getting what neurotypical individuals are communicating, which causes further misunderstandings.

> *'when the neurotypical conversation starts, the one made of subtexts <sp> of politeness <sp> of idioms <sp> I don't understand anything' (P02, diagnosed autistic woman, aged 52)*

Missing information

Participants noticed differences in neurotypical people's communication in terms of informativeness. Autistic people usually specify exhaustively what they are saying and dislike undefined or uncertain information, whilst neurotypical people seem to overlook certain aspects, such as details or topic of a meeting.

> *'neurotypicals usually tend to <eeh> say things they won't do, I mean, because it's something you say <sp> […] that kind of 'yeah, we'll see, we'll organise <ooo>or this one we'll do' <sp> I need to know<ww> when, how and*

> *why <sp> something that I […] get more of a match on in communication with neuroatypical/neurodivergent people' (P12, diagnosed autistic woman, aged 31)*

However, this characteristic could vary depending on contacts with autistic people. It seems that neurotypical parents of autistic people try to understand more and tend to develop a communicative style that is more complete and therefore comprehensible to autistic people. As P20 (diagnosed autistic woman, aged 40) states: 'that's probably just a matter of education'.

Small talk

This characteristic seems to be predominant in neurotypical communication, that is polite conversation on unimportant matters. Participants perceived it as a part of the structure of neurotypical interaction in terms of conversational content, which feels particularly light and without purpose or meaning. For instance, asking out of mere politeness how the person is feeling or proposing to organise something without any actual will.

> *'so, talking about meaningless things to fill the silence doesn't <eeh> it doesn't belong to me <sp> and apparently it's the national sport […] of neurotypical people' (P20, diagnosed autistic woman, aged 40)*

From the autistic participants' perspective, small talk is meaningless and shallow, indeed it does not communicate anything important, like talking about the current weather. Autistic people prefer investing time and energy in more interesting conversations, in which they can exchange some form of knowledge.

Neurotypical scripts

Participants reported that neurotypical communication seems ciphered and scripted, following certain rules and procedures to conduct conversations, regardless of their content, restraining neurotypical people regarding how much they can express themselves in each specific context. It is perceived as a very stereotyped and strict interaction including a set of predetermined words to use or avoid in specific situations, and rules that cannot be broken in order to not be perceived as socially awkward. As they include all the social conventions and codes governing social interactions, these can result in indirect communication, missing information and small talk, as well as in reprehension for autistic people who do not follow them. Some participants recalled how they could not understand and adopt these scripts, and they were reprimanded for not adapting to the neurotypical world.

> *'instead, let's say neurotypical communication tends to be more reserved, kind of, and even more <sp> from my point of view, also more stereotyped' (P01, diagnosed autistic gender-questioning person, aged 34)*

Neurotypical masking

This category is related to the previous one. However, it has its own specificity as it reports on the idea that neurotypical communication, as characterised by the other categories in terms of directness, (light) content and (stereotyped) way of interacting, can sometimes be the result of a filter, of a mask, that resembles autistic masking, and was therefore named by some of them as 'masking neurotipico'. Yet, this filter seems to mask thoughts and, particularly, emotions, more than behaviours. Autistic people perceive this mask, worn by neurotypical people, as a part of the aforementioned structure that seems to rule neurotypical communication, but potentially harmful to neurotypical people as well.

> *'the mask of neurotypicals […] that is <sp> when […] a person in front of you […] says certain things, but with body language is showing anything but that […] this way, you're preventing me from replying correctly, because I should reply on a level that, however, is not the real one […] […] I would have to unmask you, to violently force you <sp> truly creating a rupture<hh> so that you can break that mask' (P05, diagnosed autistic gender-questioning person, aged 38)*

Interpersonal communication

There seem to be differences in autistic perception of communication depending on the interlocutor's neurotype. Ease (and discomfort) can be influenced by neurotype-matching. This is represented in the themes *Autistic–Neurotypical Communication* and *Autistic–Autistic Communication* (Figure 2).

Autistic–Neurotypical communication

This theme addresses the autistic perspective on cross-neurotype communication. This includes difficulties with conversational turns and anger because of misinterpre-

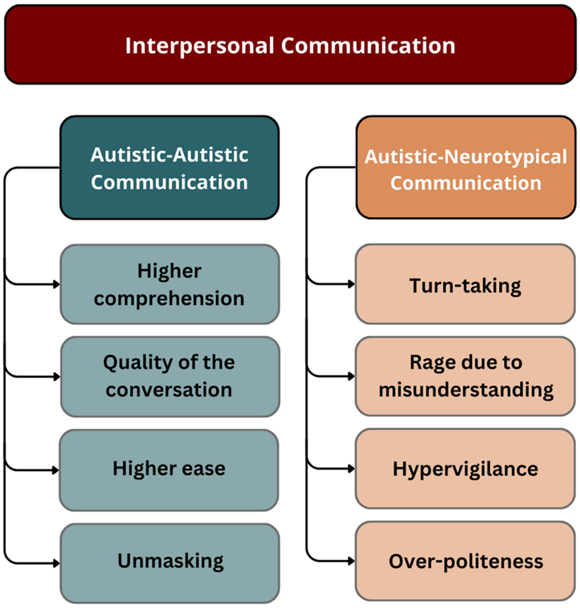

FIGURE 2 Themes and categories in interpersonal communication.

tation of their message by neurotypical people, but also hypervigilance and over-politeness.

Turn-taking

Participants explained their perception of being incapable of performing conversational turns the way neurotypical people do, both in one-to-one and group conversations. Indeed, autistic people often interrupt neurotypical ones, or they do not talk at all, which is perceived as rude or presumptuous by neurotypical people. Conversely, autistic individuals talk over each other and even have long conversational turns, as seen in *Autistic monologue* (see *Autistic Communication*), without taking offence, waiting for the interlocutor's monologue to end to start their own.

> *'I've always had this feeling <sp> as if the typical conversation was a <eeh> a dance, I mean, a choreography of which I didn't know the steps, and so, in one way or another, I stepped on someone's feet <sp> either because I said something wrong or because I didn't know what to answer' (P10, diagnosed autistic non-binary person, aged 28)*

Rage due to misunderstanding

This theme explores the anger felt by participants when misunderstood by neurotypical people, notwithstanding their continuous effort to explain their perspective. This is described as painful as it happens very frequently, making it seem a one-way communication. Moreover, this also occurs during high-emotional impact conversations, possibly making it more painful.

Participants ascribe this misinterpretation to an inability of neurotypical people to conceive anything different from their communicative style. As such, neurotypical people would be unable to understand autistic people as they do not communicate in the normative neurotypical way. Participants reported feeling enraged because neurotypical people do not seem to try to understand their differences and continue reading autistic behaviour through neurotypical schemes.

> *'if I <sp> who spent a life being misunderstood by others, now that I try to explain to you why you couldn't understand me<ee> no, you have to talk over me and tell me<ee> make me<ee> do, like, neurotypical-splaining <laugh> I have no idea how to call it <laugh> <eeeh> so, I really see a<aa> difficulty in conceiving that a different communicative style could exist <sp> so, there's actually a difficulty' (P08, diagnosed autistic woman, aged 31)*

Hypervigilance

When talking with neurotypical people, participants feel the need to be constantly aware of their words and body language, whilst this does not happen when conversing with autistic peers. In fact, whilst they learnt masking at such an early age that it became an unconscious mechanism regardless of the interlocutor's neurotype, they perceive a particular level of activation with neurotypical people. This resembles some experiences and constructs from academic literature on anxiety, such as continuous scanning of the environment (and themselves) for potential threats. It seems to be linked to the previous category: participants try to control and moderate their reactions, fearing they might be miscomprehended. This high level of self-consciousness is based on a conscious, energy-demanding, strategy of meticulous observation of their neurotypical interlocutor.

> *'during conversations with neurotypical people <sp> it's as if I had to think ten thousand things <sp> because I'm masking <sp> because I don't understand what's going on and I've to decode a ton <sp> and I really can't, or I make a great effort to… focus on the content <sp> and also having appropriate reactions because <sp> I don't have enough mental space to understand how I feel' (P10, diagnosed autistic non-binary person, aged 28)*

Over-politeness
Autistic individuals tend to be overly courteous with neurotypical people. Participants explained they developed over-politeness and accommodative strategies because they were reprimanded in childhood for being 'rude'. Moreover, realising their communication is too direct, and could be perceived as unkind, autistic people try to compensate by apologising for it. This category is also linked to *Hypervigilance*: autistic people are overly alert in choosing words, to avoid being perceived as impolite.

> *'It also happens to me<ee> almost always to be too polite <sp> I mostly think it's because I can't measure <sp> the correct dose of courtesy and assertiveness <sp> and I've found out, due to past experiences, that the safest thing to do it's being too polite instead of too assertive' (P22, diagnosed autistic woman, aged 27)*

Autistic–Autistic communication

This theme regards interactions between autistic people. Participants described conversations with autistic peers as characterised by higher comprehension and quality, along with comfort and unmasking.

Higher comprehension
Participants experienced more understanding from autistic peers than neurotypical people. This includes precise and effective communication, characterised by directness, which is perceived as helpful to avoid misinterpretations, rather than rude. This holds even in group conversations, where participants felt more engaged than in neurotypical groups, due to differences in turn-taking and fluid topic changes. Both seemingly-unrelated links to other subjects and talking over someone are interpreted as difficulties to contain the excitement about a topic. Sharing these differences from the norm allows them to better comprehend each other.

> *'we talk a lot, for hours, of the same topic and keep on talking over one another <sp> but always with some sort of checking so that actually <sp> nobody feels, feels that they can't speak, or feels that the other person is intruding their conversational <sp> turn <sp> there's a sort of structure anyway <sp> though it's a fundamentally different structure than the neurotypical' (P10, diagnosed autistic non-binary person, aged 28)*

Quality of the conversation
According to participants, talking with autistic peers increases conversational quality in terms of interest, in-depth analysis, emotional involvement, clarity and logic. Autistic people love giving details, to allow their interlocutor to follow their reasoning. Participants also report they would love receiving such in-depth communication to better understand what others say. Overall, autistic communication seems to be longer, given the amount of details provided.

> *'the clarity and the fact <sp> of being very direct <sp> concerning [neuro]divergent people, it's so evident to me <sp> but <sp> I notice also more presence <sp> of logic inside the discourse' (P19, diagnosed autistic woman, aged 19)*

Higher ease
Participants feel more comfortable conversing with autistic peers, for several reasons. They can be direct without being hypervigilant, achieving clarity and less misinterpretations; they can focus on what is said instead of spending much energy to control their words and body. Not respecting turn-taking is not perceived as impolite, thus participants feel like interacting freely. Also, unmasking feels easier (see *Unmasking*). All these make the conversation flow more pleasantly.

> *'while talking, talking with non-neurotypical people <sp> we're dancing together <sp> and it's so immediate and fluid that it's beautiful' (P10, diagnosed autistic non-binary person, aged 28)*

Unmasking
Unmasking is possible with autistic peers or long-lasting friends, with whom they feel comfortable—even with neurotypical ones who have learnt how to relate to their autistic friends. Unmasking is also usually easier with children alone, for example, as teachers in a class, because they know that their truest selves will be accepted instead of judged. Coming back home and being able to unmask with loved ones is described as a relief, liberatory and fulfilling:

> *'compared to the past, I see how my real personality emerges more compared to all the masks I've built <eeh> which is why I didn't feel like myself, even with people close to me' (P01, diagnosed autistic gender-questioning person, 34 years old)*

International Journal of Language & Communication Disorders

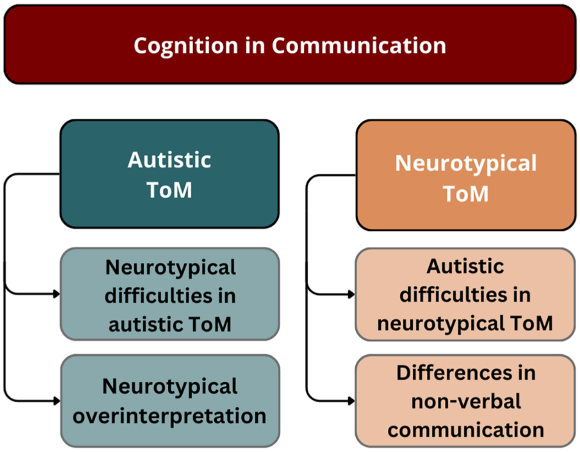

FIGURE 3 Themes and categories in cognition in communication.
Abbreviation: ToM, Theory of Mind.

Cognition in communication

The analysis revealed differences in cognition between autistic and neurotypical people as perceived by participants. It seems that autistic people experience challenges in *Neurotypical ToM*, but also that neurotypical people find understanding *Autistic ToM* difficult (Figure 3).

Neurotypical ToM

This theme considers understanding neurotypical intentions from the autistic perspective. This includes general difficulties in cognitive empathy towards neurotypical people and specific struggles in comprehending neurotypical non-verbal communication.

Autistic difficulties in neurotypical ToM
Participants shared their difficulties with implicit messages delivered through irony and sarcasm, and their uncertainty in inferring neurotypical individuals' mental states. They try to guess what neurotypical people want to convey and ask questions to grasp it. This is often described as an active and conscious deduction process based on various verbal and non-verbal cues they have read about.

Some participants suggested the experience is similar to that of people from other cultures, speaking different languages, using different writing systems. They felt the need to educate themselves on (neurotypical) communication and social functioning the way they would learn about a different language or culture—or, even, species. However, studying it does not result in immediate understanding. Some reported studying manuals, taking notes, creating mental databases of sentences and yet, still feeling 'like a fortune-teller' or 'rolling the dice', interpreting neurotypical behaviour. Even after years of studying it, neurotypical 'foreign' language remains obscure for autistic people.

Some described this as an absence of synchronicity or having a barrier between oneself and neurotypical people, existing in both directions: neither the autistic nor the neurotypical person comprehends the other. It feels impossible to establish real connections, although they can see each other:

> *'it's as if I had a glass between me and other people <sp> they see me, and I see them, but I know that there's the glass and we can't touch [...] it's as if my words would approximately go in their direction and theirs approximately in mine, but in reality there's no sync' (P15, diagnosed autistic woman, aged 34)*

Differences in non-verbal communication
Participants reported struggles with neurotypical non-verbal communication. Although they learned (often through studying and courses) what is socially perceived as the correct non-verbal reaction (e.g., facial expression or voice inflection), they sometimes find it difficult to perform in specific situations. These difficulties are reflected both in understanding neurotypical people and in using non-verbal hints improperly according to neurotypical people, regardless of readings and courses on proxemics.

> *'my colleagues have stated more than once that I'm robotic, I'm analytical <sp> I'm schematised and all that, I frequently don't use the voice inflection they deem correct' (P08, diagnosed autistic woman, aged 31)*

Autistic ToM

This theme explores autistic participants' perceptions of being misunderstood by neurotypical people, who seem to be unable to deduce autistic ToM, mostly leading to overinterpretation. This means neurotypical people would present a deficit in cognitive empathy towards autistic people.

Neurotypical difficulties in autistic ToM
Participants reported that neurotypical individuals have incredible difficulties in comprehending them. From their perspective, neurotypical people try to fit them in their mental patterns without trying to understand them. Indeed, participants felt like, even in the rare genuine attempts at comprehending them, neurotypical individuals would still eventually return to their neuro-normative

mental schemas to decipher reality. Participants also reported their way of expressing emotions is misunderstood whenever it does not reflect the neurotypical way, and being criticised and asked to change both for being too revealing (which is perceived as aggressive) and for not displaying enough of their emotions (usually misinterpreted as a lack of emotion, or as a sense of superiority). Moreover, neurotypical individuals seem to demand a quick reaction, without contemplating or inferring that autistic people might need more time for processing information and producing an adequate response. These reactions are also usually misinterpreted. It also appears that neurotypical people either misunderstand as a lack of knowledge or even as an attack, when autistic people ask questions to merely increase their understanding of the topic at hand. From the participants' perspective, neurotypical people have difficulties in conceiving different communicative styles, possibly because being part of the majority made them believe no comprehension effort would ever be needed from their part:

> *'I've always had the feeling that other people would try to <ehm> insert me in their, their pre-made mould <sp> that is, those were the moulds they had <sp> I had to remain stuck inside it, but I didn't / didn't fit at all' (P05, diagnosed autistic gender-questioning person, 38 years old)*

Neurotypical overinterpretation

Participants described the attempt of neurotypical people to understand them through their cognitive system, adopting their mental patterns, leads them to misunderstand autistic intentions in a specific way, most of the time: they tend to overinterpret what autistic people say. This seems related to their tendency to frequently deliver implicit messages, which autistic people do not have. Thus, neurotypical people always look for something they think would be implied and eventually tend to fabricate hidden meanings into what autistic individuals say.

Sometimes this tendency is perceived by autistic individuals as violent: they feel forced to accept an implied meaning they had not meant. This seems to be related to the autistic tendency of using fewer facial expressions or different non-verbal communication (e.g., stimming or looking elsewhere to pay attention) than neurotypical people. Consequently, neurotypical people perceive them as angry, uninterested and rude, when they were not:

> *'also, in interpersonal relations <sp> like, they almost tried to <sp> like, to impose their perception on me <sp> in other words, to make it mine, you know? <sp> I mean, I don't know <eh> 'but you're angry with me' <sp> <eh> 'but you're not really interested in this thing' <sp> but who told you?' (P18, diagnosed autistic man, 47 years old)*

Overinterpretation seems to arise because neurotypical people place great importance not only on the utterances, but also on proxemics and body language, which is used differently by (and have different meanings to) autistic and neurotypical individuals. Autistic people feel they do not convey enough messages through non-verbal communication according to the neurotypical norm, and so neurotypical interlocutors fill the perceived gap with their overinterpretation.

DISCUSSION

Overall, our modifications of VFGDs to accommodate autistic participants seemed to provide a safe ground for a productive moderated interaction for participants, who participated actively, using all means of expression and accommodation, and gave positive feedback. Participants' feedback was in line with the neurotypical assistants' remarks and field notes, which mentioned most themes later developed in the analysis. Mutual reflexivity was important to mitigate potential biases an entirely neurotypical or autistic perspective on neurodiverse interactions might introduce.

Focus groups entail a risk of conformance within the group and censorship of minoritised ideas (Carey & Smith, 1994); however, some participants expressed disagreement on some points, and many expressed ideas nobody else mentioned.

Taken together, the results indicate that participants identified several characteristics of a neurotypical communication style and their own. They also reported difficulties in communication both regardless of the interlocutor's neurotype and specific to conversations with neurotypical people, mainly due to differences in mentalising, which do not apply to autistic peers.

Interestingly, participants highlighted features of neurotypical communication they perceive as dysfunctional to successful communication. Most importantly, the overarching theme of *Cognition in Communication* suggests that autistic difficulties in neurotypical ToM are mirrored by neurotypical difficulties in autistic ToM.

Most themes are in line with previous literature. In fact, *Autistic Communication* reports, through a more neutral framing, known autistic characteristics (or even diagnostic criteria): directness (usually mentioned as bluntness or impoliteness); monologuing (usually framed as a turn-taking issue) and special interests, perceived energy demand and sensory interferences (usually referred to as sensory hyper/hypo-reactivity). This suggests that group dynamics and researchers' reflexivity seem to have avoided the potential censorship of any view that could frame some difficulties as inherently part of autistic functioning, as they were presented as existing regardless of the interlocutor's neurotype.

Masking and camouflaging are also known in the literature, particularly in women (Cook et al., 2021), predominant in our sample. Camouflaging was perceived as present regardless of the interlocutor's neurotype. However, this was particularly important in *Autistic–Neurotypical Communication*. In fact, participants also reported hypervigilance and overpoliteness in conversation with neurotypical people, which might be forms of self-monitoring and self-policing ascribable to camouflaging. Moreover, these interactions make autistic people frame their monologuing as a turn-taking problem and trigger their rage due to misunderstanding.

Findings on *Neurotypical Communication* and *Neurotypical ToM* also confirm previous literature: participants described neurotypical language as characterised by indirectness and scripts, as well as small talk (as pragmatics literature testifies). Interestingly, participants also mentioned that neurotypical people could be masking themselves: they would hide or even lie about their feelings, potentially resulting in harm to themselves in addition to interpretation difficulties. This observation paves the way to the two newer findings: that this does not happen in *Autistic–Autistic Communication*, and that autistic people might have a different ToM.

Communication within the spectrum

Autistic–Autistic Communication was reported as characterised by higher levels of comprehension, quality of the conversation even on other people's special interests and higher ease of communication, facilitating unmasking.

This confirms previous studies highlighting information transfer between autistic people is more successful than between an autistic and a non-autistic person (Crompton et al., 2020b) and that rapport and flow seem to increase in autistic–autistic as compared to cross-neurotype communication, so much that individual communicative competence seem to improve (Williams et al., 2021; Crompton et al., 2020c). It also corroborates pragmatic implications of the Double Empathy problem (Milton, 2012).

Evidence in favour of a double empathy

Within the overarching theme *Cognition in Communication*, participants confirmed their difficulties understanding neurotypical people, but they also reported neurotypical difficulties in comprehending their ToM. Notably, a specific type of neurotypical interpretative behaviour was developed as a distinct category: autistic people reported that neurotypical people tend to overinterpret their statements, particularly related to the autistic challenge to derive implications from what is said by neurotypical interlocutors. This could reflect neurotypicals' indirect communication expectation based on neurotypical ToM, even when autistic individuals communicate directly.

This analysis supports the view that autistic comprehension difficulties stem from mutual misunderstanding during cross-neurotype interactions (Milton 2012; Williams et al., 2021). Further, it suggests it might be useful to reframe ToM and, possibly, to introduce the construct of autistic ToM. In fact, from an autistic perspective, neurotypical people seem to lack sufficient autistic ToM abilities. These findings suggest that using a methodology that challenges the usual power dynamics influencing the social construction of knowledge can highlight often disregarded personal accounts, contributing to the awareness of new perspectives and potentially challenge the socially accepted truth.

It is also possible to speculate that, if adequately informed on neurotypical-autistic ToM differences, neurotypical abilities in autistic ToM would improve. Recent studies on autism acceptance training seem to confirm this hypothesis: Jones et al. (2021) found that neurotypical and autistic people interacting in dyads expressed greater interest in hanging out again if the neurotypical person had attended the training. This perspective change on interaction as co-constructed prompts unexplored conceptualisations of 'interventions' targeting cross-neurotype misunderstanding, rather than autistic skills exclusively.

Limitations and future studies

Our study has several limitations, particularly with regard to transferability. In fact, its results are rooted in a specific range of experiences within the autism spectrum, reflecting the intersection of White, Italian, high-masking, late-diagnosed autistic identities of the participants. It is still difficult for ethnic minorities to get diagnosed in Italy, and including self-identified autistic participants was not enough to diversify our sample. Autistic men's and low-masking perspectives were also less represented. Future studies would benefit from purposeful sampling techniques to ensure greater diversity.

Furthermore, our participants' views on masking and politeness could be influenced by their late diagnosis: they recalled how their 'inappropriate behaviors' were reprimanded and how they changed accordingly, not knowing their neurotype. However, this kind of repression could also occur with early diagnosis, sometimes through clinical intervention. Further studies should investigate how these results could partially be transferred to autistic people with intellectual disability and/or linguistic difficulties.

Lastly, quantitative studies could test whether these themes are part of the autistic experience on a broader level.

Implications for speech and language therapy

Insights from this study carry implications for clinical education and speech and language therapy (SLT) practice. In line with previous studies (Cummins, Pellicano & Crane, 2020), our results suggest both individual support and broader societal changes are needed to make communication a better experience for autistic people. Preferred modes of communication and co-occurrence of linguistic difficulties have to be considered in SLT practice, as no prescriptive approach can work for any autistic individual. However, integrating these findings into SLT practice could empower autistic people, letting them know they could find comprehension in the autistic community, and that communication problems are mutual misunderstandings to work on collaboratively. Similar approaches have been well received in other populations (e.g., people who stammer: Beilby et al., 2012; Cheasman et al. 2015), and would probably ease masking-related mental health difficulties.

SLTs could also participate in broader societal changes, fostering awareness on the Double Empathy Problem and engaging with autistic people through participatory approaches. This could create positive environments for autistic people and their families, enhance neurotypical individuals' awareness of their limits in conversations with autistic people and of their role in mutual comprehension.

CONCLUSIONS

Results indicated that participants experienced difficulties in communication that were rooted in mutual misunderstanding and in a perceived difference between the neurotypical and autistic ToM. Further studies evaluating this phenomenon from a quantitative point of view are needed. Nevertheless, awareness of these experiences might result in reduced misunderstandings in cross-neurotype communication.

ACKNOWLEDGEMENTS

The authors would like to thank Lorenzo Scalchi and Francesco Fattori from 'Codici Ricerca e Intervento' for their support as collaborators in the VFGDs and the helpful comments in the materials preparation phase. The authors are also thankful to the participants for the generous and insightful answers they provided to our questions on such a delicate and under-researched topic. This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The dataset is available on the Open Science Framework (https://osf.io/p8ubr).

PARTICIPANTS CONSENT STATEMENT

Written and signed informed consent was obtained from all of the participants.

ORCID

Eleonora Marocchini https://orcid.org/0000-0003-4970-4511

Irene Baldin https://orcid.org/0000-0002-1136-8324

ENDNOTES

[1] Of these four non-binary people, three declared their gender identity does not correspond to the sex they have been assigned at birth and one preferred not to answer the question; of the two people tagged as gender-questioning, one defined themself as gender-questioning and preferred not to answer the following question, and one defined herself as a human person, though accepting woman as a label, and declared her gender identity only partially corresponds to the sex she was assigned at birth.

[2] However, Zoom is still preferable to other video-conferencing platforms due to its ease of use, cost-effectiveness, data management and security features (see Archibald et al., 2019).

REFERENCES

American Psychiatric Association (APA). (2022) Neurodevelopmental disorders. In Diagnostic and statistical manual of mental disorders (Fifth edition, Text revision).

Archibald, M.M., Ambagtsheer, R.C., Casey, M.G., & Lawless, M. (2019) Using zoom videoconferencing for qualitative data collection: perceptions and experiences of researchers and participants. *International Journal of Qualitative Methods*, 18, 1–8. https://doi.org/10.1177/1609406919874596

Baron-Cohen, S., Leslie, A., & Frith, U. (1985) Does the autistic child have a "theory of mind"? *Cognition*, 21, 13–125.

Baron-Cohen, S., & Wheelwright, S. (2004) The empathy quotient: an investigation of adults with Asperger syndrome or high functioning autism, and normal sex differences. *Journal of Autism and*

Developmental Disorders, 34(2), 163–175. https://doi.org/10.1023/b:jadd.0000022607.19833.00

Beilby, J.M., Byrnes, M.L., & Yaruss, J.S. (2012) Acceptance and commitment therapy for adults who stutter: psychosocial adjustment and speech fluency. *Journal of Fluency Disorders*, 37(4), 289–299. https://doi.org/10.1016/j.jfludis.2012.05.003.

Bosco, F.M., Tirassa, M., & Gabbatore, I. (2018) Why pragmatics and theory of mind do not (completely) overlap. *Frontiers in Psychology*, 9, 1453

Brewer, R., Biotti, F., Catmur, C., Press, C., Happé, F., Cook, R., & Bird, G. (2016) Can neurotypical individuals read autistic facial expressions? Atypical production of emotional facial expressions in autism spectrum disorders. *Autism Research*, 9(2), 262–271.

Brock, J., Norbury, C., Einav, S., & Nation, K. (2008) Do individuals with autism process words in context? Evidence from language-mediated eye-movements. *Cognition*, 108(3), 896–904.

Burr, V. (1995) *An introduction to social constructionism*. London: Routledge.

Carey, M.A., Smith, M. (1994) Capturing the group effect in focus groups: a special concern in analysis. *Qualitative Health Research*, 4, 123–27.

Chapman, R. (2019) Neurodiversity theory and its discontents. In Tekin, Ş. & Bluhm, R. (Eds.), *The bloomsbury companion to philosophy of psychiatry*. New York: Bloomsbury Academic.

Cheasman, C., Simpson, S., & Everard, R. (2015) Acceptance and speech work: the challenge. *Procedia—Social and Behavioral Sciences*, 193, 72–81. https://doi.org/10.1016/j.sbspro.2015.03.246

Chevallier, C. (2012) Theory of mind and autism: beyond Baron-Cohen et al.'s Sally-Anne study. In Slater, A.M. & Quinn, P.C. (Eds.), *Developmental psychology: revisiting the classic studies*. Thousand Oaks: Sage, pp. 148–163.

Chevallier, C., Wilson, D., Happé, F., & Noveck, I. (2010) Scalar inferences in autism spectrum disorders. *Journal of Autism and Developmental Disorders*, 40(9), 1104–1117.

Cook, J., Hull, L., Crane, L., & Mandy, W. (2021) Camouflaging in autism: a systematic review. *Clinical Psychology Review*, 89, 102080. https://doi.org/10.1016/j.cpr.2021.102080

Corbett, B.A., Schwartzman, J.M., Libsack, E.J., Muscatello, R.A., Lerner, M.D., Simmons, G.L., & White, S.W. (2021) Camouflaging in autism: examining sex-based and compensatory models in social cognition and communication. *Autism Research*, 14, 127–142. https://doi.org/10.1002/aur.2440

Crompton, C.J., Hallett, S., Ropar, D., Flynn, E., & Fletcher-Watson, S. (2020a) 'I never realised everybody felt as happy as I do when I am around autistic people': a thematic analysis of autistic adults' relationships with autistic and neurotypical friends and family. *Autism*, 24(6), 1438–1448.

Crompton, C.J., Ropar, D., Evans-Williams, C.V., Flynn, E.G., & Fletcher-Watson, S. (2020b) Autistic peer-to-peer information transfer is highly effective. *Autism*, 24(7), 1704–1712.

Crompton, C.J., Sharp, M., Axbey, H., Fletcher-Watson, S., Flynn, E.G., & Ropar, D. (2020c) Neurotype-matching, but not being autistic, influences self and observer ratings of interpersonal rapport. *Frontiers in Psychology*, 11, 586171. https://doi.org/10.3389/fpsyg.2020.586171

Cummings, L. (2009) *Clinical pragmatics*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511581601

Cummins, C., Pellicano, E., & Crane, L. (2020) Autistic adults' views of their communication skills and needs. *International Journal of Language & Communication Disorders*, 55(5), 678–689. https://doi.org/10.1111/1460-6984.12552

Dawson, G., Franz, L., Brandsen, S. (2022) At a crossroads-reconsidering the goals of autism early behavioral intervention from a neurodiversity perspective. *JAMA Pediatrics*, 176(9), 839-840. 10.1001/jamapediatrics.2022.2299

Edey, R., Cook, J., Brewer, R., Johnson, M.H., Bird, G., & Press, C. (2016) Interaction takes two: typical adults exhibit mindblindness towards those with autism spectrum disorder. *Journal of Abnormal Psychology*, 125(7), 879–885.

Fletcher-Watson, S., Adams, J., Brook, K., Charman, T., Crane, L., Cusack, J., et al. (2019) Making the future together: shaping autism research through meaningful participation. *Autism*, 23, 943–953. 10.1177/1362361318786721

Frith, U. (1989) A new look at language and communication in autism. *The British Journal of Disorders of Communication*, 24(2), 123–150. https://doi.org/10.3109/13682828909011952

Gernsbacher, M.A., & Pripas-Kapit, S.R. (2012) Who's missing the point? A commentary on claims that autistic persons have a specific deficit in figurative language comprehension. *Metaphor and Symbol*, 27(1), 93–105.

Gernsbacher, M.A., & Yergeau, M. (2019) Empirical failures of the claim that autistic people lack a theory of mind. *Archives of Scientific Psychology*, 7(1), 102–118.

Giorgi, A. (1997) The theory, practice, and evaluation of the phenomenological method as a qualitative research procedure. *Journal of Phenomenological Psychology*, 28, 235–260.

Graneheim, U.H. & Lundman, B. (2004) Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. *Nurse Education Today*, 24, 105–112.

Happé, F., Cook, J.L., & Bird, G. (2017) The structure of social cognition: in(ter)dependence of sociocognitive processes. *Annual Review of Psychology*, 68(1), 243–267.

Happé, F. (1995) The role of age and verbal ability in the theory of mind task performance of subjects with autism. *Child Development*, 66(3), 843–855.

Happé, F. (1993) Communicative competence and theory of mind in autism: a test of relevance theory. *Cognition*, 48(2), 101–119.

Jones, D.R., Morrison, K.E., DeBrabander, K.M., Ackerman, R.A., Pinkham, A.E., & Sasson, N.J. (2021) Greater social interest between autistic and non-autistic conversation partners following autism acceptance training for non-autistic people. *Frontiers in Psychology*, 12, 739147. doi: 10.3389/fpsyg.2021.739147

Kanske, P., (2018) The social mind: disentangling affective and cognitive routes to understanding others. *Interdisciplinary Science Reviews*, 43, 115–124. https://doi.org/10.1080/03080188.2018.1453243.

Kanske, P., Böckler, A., Trautwein, F.-M., Parianen Lesemann, F.H., Singer, T., (2016) Arestrong empathizers better mentalizers? Evidence for independence and interaction between the routes of social cognition. *Social Cognitive and Affective Neuroscience*, 11, 1383–1392. https://doi.org/10.1093/scan/nsw052

Leung, L. (2015) Validity, reliability, and generalizability in qualitative research. *Journal of Family Medicine and Primary Care*, 4(3), 324-327

Livingston, L.A., Carr, B., & Shah, P. (2019) Recent advances and new directions in measuring theory of mind in autistic adults. *Journal of Autism and Developmental Disorders*, 49(4), 1738–1744. https://doi.org/10.1007/s10803-018-3823-3

Loukusa, S., Gabbatore, I., Kotila, A.R., Dindar, K., Mäkinen, L., Leinonen, E. et al. (2023) Non-linguistic comprehension, social inference and empathizing skills in autistic young adults, young adults with autistic traits and control young adults: group differences and interrelatedness of skills. *International Journal of Language & Communication Disorders*, 58, 1133–1147. https://doi.org/10.1111/1460-6984.12848

Marocchini, E. (2023) Impairment or difference? The case of Theory of mind abilities and pragmatic competence in the autism spectrum. *Applied Psycholinguistics*, 44(3), 365–383. doi:10.1017/S0142716423000024

Marocchini, E., Di Paola, S., Mazzaggio, G., & Domaneschi, F. (2022) Understanding indirect requests for information in high-functioning autism. *Cognitive Processing*, 23, 129–153. https://doi.org/10.1007/s10339-021-01056-z

Milton, D. (2012) On the ontological status of autism: the double empathy problem. *Disability & Society*, 27(6), 883–887.

Morgan, D.J. (1996) Focus Groups. *Annual Review of Sociology*, 22(1), 129–152

Olderbak, S., Wilhelm, O., Olaru, G., Geiger, M., Brenneman, M.W., & Roberts, R.D. (2015) A psychometric analysis of the reading the mind in the eyes test: toward a brief form for research and applied settings. *Frontiers in Psychology*, 6, 1503. https://doi.org/10.3389/fpsyg.2015.01503

Ozonoff, S., & Miller, J.N. (1996) An exploration of right-hemisphere contributions to the pragmatic impairments of autism. *Brain and Language*, 52(3), 411–434.

Patton, M.Q. (1999) Enhancing the quality and credibility of qualitative analysis. *Health Services Research*, 34, 1189–208

Paul, R., & Cohen, D.J. (1985) Comprehension of indirect requests in adults with autistic disorders and mental retardation. *Journal of Speech, Language, and Hearing Research*, 28(4), 475–479.

Pietkiewicz, I., & Smith, J.A. (2014) A practical guide to using interpretative phenomenological analysis in qualitative research psychology. *Psychological journal*, 20(1), 7–14.

Premack, D., & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4), 515–526. https://doi.org/10.1017/s0140525x00076512

Quesque, F., & Rossetti, Y. (2020) What do theory-of-mind tasks actually measure? Theory and practice. *Perspectives on Psychological Science*, 15(2), 384–396. https://doi.org/10.1177/1745691619896607

Rees, C.E., Crampton, P.E.S., Monrouxe, L.V. (2020) Re-visioning academic medicine through a constructionist lens. *Academic Medicine: Journal of the Association of American Medical Colleges*, 95(6), 846-850. 10.1097/ACM.0000000000003109

Savy, R. (2007) *Specifiche per la trascrizione ortografica annotata dei testi*. In Albano Leoni, F. & Giordano, R. (Eds.), *Italiano Parlato. Analisi di un dialogo*. Napoli: Liguori.

Schaafsma, S.M., Pfaff, D.W., Spunt, R.P., & Adolphs, R. (2015) Deconstructing and reconstructing theory of mind. *Trends in Cognitive Sciences*, 19(2), 65–72. https://doi.org/10.1016/j.tics.2014.11.007

Scheeren, A.M., de Rosnay, M., Koot, H.M., & Begeer, S. (2013) Rethinking theory of mind in high-functioning autism spectrum disorder. *Journal of Child Psychology and Psychiatry*, 54(6), 628–635.

Seuren, L.M., Wherton, J., Greenhalgh, T., & Shaw, S.E. (2021) Whose turn is it anyway? Latency and the organization of turn-taking in video-mediated interaction. *Journal of Pragmatics*, 172, 63–78. https://doi.org/10.1016/j.pragma.2020.11.005

Sirota, K.G. (2004) Positive politeness as discourse process: politeness practices of high-functioning children with autism and Asperger Syndrome. *Discourse Studies*, 6(2), 229–251. https://doi.org/10.1177/1461445604041769

Sperber, D., & Wilson, D. (2002) Pragmatics, modularity and mind-reading. *Mind & Language*, 17, 3–23. https://doi.org/10.1111/1468-0017.00186

Stietz, J., Jauk, E., Krach, S., Kanske, P., (2019) Dissociating empathy from perspective-taking: evidence from intra- and inter-individual differences research. *Front. Psychiatry*, 10, 126. https://doi.org/10.3389/fpsyt.2019.00126

Surian, L. (1996) Are children with autism deaf to Gricean maxims? *Cognitive Neuropsychiatry*, 1(1), 55–72.

Tong, A., Sainsbury, P., Craig, J. (2007) Consolidated criteria for reporting qualitative research (COREQ): a 32-item checklist for interviews and focus groups. *International Journal for Quality in Health Care*, 19(6), 349-357

Tuttas, C.A. (2015) Lessons learned using Web conference technology for online focus group interviews. *Qualitative Health Research*, 25(1), 122–133. https://doi.org/10.1177/1049732314549602

Vaismoradi, M., Turunen, H., Bondas, T. (2013) Content analysis and thematic analysis: implications for conducting a qualitative descriptive study. *Nursing & Health Sciences*, 15(3), 398-405

van Tiel, B., Deliens, G., Geelhand, P., Murillo Oosterwijk, A., & Kissine, M. (2021) Strategic deception in adults with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 51, 255–266. https://doi.org/10.1007/s10803-020-04525-0

Volden, J., & Sorenson, A. (2009) Bossy and nice requests: varying language register in speakers with autism spectrum disorder. *Journal of Communication Disorders*, 42, 58–73. https://doi.org/10.1016/j.jcomdis.2008.08.003

White, R. (2004) Discourse analysis and social constructionism. *Nurse Researcher*, 12(2), 7-16. doi: 10.7748/nr.12.2.7.s3.

Williams, G.L., Wharton, T., & Jagoe, C. (2021) Mutual (Mis)understanding: reframing autistic pragmatic "impairments" using relevance theory. *Frontiers in Psychology*, 180, 121–130.

Yang, C., Liu, D., Yang, Q., Liu, Z., & Prud'hommeaux, E. (2021) Predicting pragmatic discourse features in the language of adults with autism spectrum disorder. *ACL Anthology*, 284–291. https://doi.org/10.18653/v1/2021.acl-srw.29

Ying Sng, C., Carter, M., & Stephenson, J. (2018). A systematic review of the comparative pragmatic differences in conversational skills of individuals with autism. *Autism & Developmental Language Impairments*, 3, 1–24. https://doi.org/10.1177/2396941518803806

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Marocchini, E. & Baldin, I. (2024) Cross-neurotype communication from an autistic point of view: insights on autistic Theory of Mind from a focus group study. *International Journal of Language & Communication Disorders*, 59, 2465–2482.
https://doi.org/10.1111/1460-6984.13095